We create chemistry

# Safety data sheet



BASF Safety data sheet according to the United Nations' Globally Harmonized System (UN GHS)
Date / Revised: 10.10.2023 Version: 2.0
Product: **R210 Catalyst**
(ID no. 30720401/SDS_GEN_00/EN)
Date of print 03.07.2025

## 1. Identification

### Product identifier

## R210 Catalyst

### Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: Intermediate
Recommended use: Intermediate (isolated and transported)

### Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Petrochemicals

Telephone: +49 621 60-42151
E-mail address: sds-petrochemicals@basf.com

### Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

## 2. Hazards Identification

### Classification of the substance or mixture

According to UN GHS criteria

Acute Tox. 5 (oral)
Eye Dam./Irrit. 2A

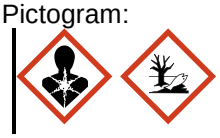
Resp. Sens. 1
Skin Sens. 1
Muta. 2
Carc. 1B
Repr. 1B (fertility)
Repr. 2 (unborn child)
Aquatic Acute 2
Aquatic Chronic 2

For the classifications not written out in full in this section the full text can be found in section 16.

## Label elements

Globally Harmonized System (GHS)

Pictogram:



Signal Word:
Danger

Hazard Statement:

| | |
|---|---|
| H319 | Causes serious eye irritation. |
| H303 | May be harmful if swallowed. |
| H334 | May cause allergy or asthma symptoms or breathing difficulties if inhaled. |
| H317 | May cause an allergic skin reaction. |
| H350 | May cause cancer. |
| H341 | Suspected of causing genetic defects. |
| H360 | May damage fertility. Suspected of damaging the unborn child. |
| H401 | Toxic to aquatic life. |
| H411 | Toxic to aquatic life with long lasting effects. |

Precautionary Statements (Prevention):

| | |
|---|---|
| P280 | Wear protective gloves, protective clothing and eye protection or face protection. |
| P201 | Obtain special instructions before use. |
| P261 | Avoid breathing dust or fume. |
| P273 | Avoid release to the environment. |
| P280 | Wear eye protection. |
| P202 | Do not handle until all safety precautions have been read and understood. |
| P284 | In case of inadequate ventilation wear respiratory protection. |
| P272 | Contaminated work clothing should not be allowed out of the workplace. |
| P264 | Wash contaminated body parts thoroughly after handling. |

Precautionary Statements (Response):

| | |
|---|---|
| P305 + P351 + P338 | IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. |
| P304 + P340 | IF INHALED: Remove person to fresh air and keep comfortable for breathing. |
| P308 + P313 | IF exposed or concerned: Get medical attention. |
| P302 + P352 | IF ON SKIN: Wash with plenty of soap and water. |
| P333 + P313 | If skin irritation or rash occurs: Get medical attention. |
| P301 + P312 | IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell. |
| P362 + P364 | Take off contaminated clothing and wash it before reuse. |
| P391 | Collect spillage. |
| P337 + P313 | If eye irritation persists: Get medical attention. |

Precautionary Statements (Storage):

| | |
|---|---|
| P405 | Store locked up. |

Precautionary Statements (Disposal):

| | |
|---|---|
| P501 | Dispose of contents and container to hazardous or special waste collection point. |

Labeling of special preparations (GHS):
The following percentage of the mixture consists of components(s) with unknown hazards regarding the acute toxicity: 17 - 26 %, dermal
The following percentage of the mixture consists of components(s) with unknown hazards regarding the acute toxicity: 12 - 19 %, oral
The following percentage of the mixture consists of components(s) with unknown hazards regarding the acute toxicity: 46 - 67 %, Inhalation - dust

### Other hazards

According to UN GHS criteria

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

See section 12 - Results of PBT and vPvB assessment.

## 3. Composition/Information on Ingredients

### Substances

Not applicable

### Mixtures

Chemical nature

Preparation based on: metal oxides

Hazardous ingredients (GHS)
According to UN GHS criteria

Cobalt molybdate

| | |
|---|---|
| Content (W/W): >= 32 % - <= 44,4 % | Acute Tox. 5 (oral) |
| CAS Number: 13762-14-6 | Resp. Sens. 1 |
| EC-Number: 237-358-4 | Skin Sens. 1 |
| REACH registration number: 01-2119948547-24 | Muta. 2 |
| | Carc. 1B (by inhalation) |
| | Repr. 1B (fertility) |
| | Repr. 2 (unborn child) |
| | Aquatic Acute 2 |
| | Aquatic Chronic 2 |
| | H303, H334, H317, H341, H350, H360, H401, H411 |
| | Specific concentration limit: |
| | Carc. 1B: >= 0,01 % |

Diiron trimolybdenum dodecaoxide

| | |
|---|---|
| Content (W/W): >= 22,5 % - <= 33,8 % | Acute Tox. 5 (oral) |
| CAS Number: 13769-81-8 | H303 |
| EC-Number: 237-389-3 | |

Molybdenum trioxide

| | |
|---|---|
| Content (W/W): >= 6,9 % - <= 10,3 % | Acute Tox. 5 (oral) |
| CAS Number: 1313-27-5 | Carc. 2 |
| EC-Number: 215-204-7 | STOT SE 3 (irr. to respiratory syst.) |
| INDEX-Number: 042-001-00-9 | Eye Dam./Irrit. 2A |
| | H319, H303, H335, H351 |

Dibismuth trioxide

| | |
|---|---|
| Content (W/W): >= 4,6 % - <= 6,9 % | Acute Tox. 5 (oral) |
| CAS Number: 1304-76-3 | H303 |
| EC-Number: 215-134-7 | |

Tricobalt tetraoxide

| | |
|---|---|
| Content (W/W): >= 1,6 % - <= 2,4 % | Resp. Sens. 1B |
| CAS Number: 1308-06-1 | Aquatic Chronic 3 |
| EC-Number: 215-157-2 | H334, H412 |

Dipotassium oxide

| | |
|---|---|
| Content (W/W): >= 0 % - <= 0,14 % | Skin Corr./Irrit. 1A |
| CAS Number: 12136-45-7 | Eye Dam./Irrit. 1 |
| EC-Number: 235-227-6 | H314 |

Tungsten trioxide

Content (W/W): >= 9,2 % - <= 13,7 %
CAS Number: 1314-35-8
EC-Number: 215-231-4

Graphite
Content (W/W): >= 3,2 % - <= 4,8 %
CAS Number: 7782-42-5
EC-Number: 231-955-3
REACH registration number: 01-2119486977-12

Silicon dioxide
Content (W/W): >= 2,4 % - <= 3,6 %
CAS Number: 7631-86-9
EC-Number: 231-545-4

For the classifications not written out in full in this section the full text can be found in section 16.

## 4. First-Aid Measures

### Description of first aid measures

First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Immediately wash thoroughly with soap and water, seek medical attention.

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

### Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

Hazards: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11. (Further) symptoms and / or effects are not known so far

### Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

## 5. Fire-Fighting Measures

### Extinguishing media

Suitable extinguishing media:
foam, dry powder, water spray

Unsuitable extinguishing media for safety reasons:
water jet, carbon dioxide

Additional information:
Use extinguishing measures to suit surroundings.

If a metal fire occurs, use dry sand, dry powder for Class D or cement.

### Special hazards arising from the substance or mixture

Cool endangered containers with water-spray. See SDS section 7 - Handling and storage.

### Advice for fire-fighters

Special protective equipment:
Wear a self-contained breathing apparatus. Special protective equipment for firefighters

Further information:
Evacuate area of all unnecessary personnel. Fight fire from maximum distance.

Extend fire extinguishing measures to the surroundings.

## 6. Accidental Release Measures

Shut off or stop released substance/product under safe conditions. Avoid dispersal of dust in the air (e.g. by clearing dusty surfaces with compressed air).

Dust in sufficient concentration can result in an explosive mixture in air. Handle to minimize dusting and eliminate open flame and other sources of ignition.

### Personal precautions, protective equipment and emergency procedures

Handle in accordance with good industrial hygiene and safety practice.

Avoid all sources of ignition: heat, sparks, open flame. Use antistatic tools.

### Environmental precautions

Discharge into the environment must be avoided.

### Methods and material for containment and cleaning up

For small amounts: Dampen, pick up mechanically and dispose of.
For large amounts: Sweep/shovel up.

Collect waste in suitable containers, which can be labeled and sealed.

## 7. Handling and Storage

### Precautions for safe handling

Handle in accordance with good industrial hygiene and safety practice. Avoid all direct contact with the substance/product. Ensure thorough ventilation of stores and work areas. Change clothes immediately after contamination. Avoid the formation and deposition of dust.

Protection against fire and explosion:
The product is not an oxidizer, not self-combustible and not explosive. The substance/product is non-combustible.

### Conditions for safe storage, including any incompatibilities

Further information on storage conditions: Keep container in a well-ventilated place. The authority permits and storage regulations must be observed.

### Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

## 8. Exposure Controls/Personal Protection

### Control parameters

Components with occupational exposure limits

1308-06-1: Tricobalt tetraoxide
1313-27-5: Molybdenum trioxide
1314-35-8: Tungsten trioxide
7631-86-9: Silicon dioxide
7782-42-5: Graphite
13769-81-8: Diiron trimolybdenum dodecaoxide
13762-14-6: Cobalt molybdate
13762-14-6: Cobalt molybdate

### Exposure controls

Personal protective equipment

Respiratory protection:
Breathing protection if dusts are formed. Particle filter with high efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P3 or FFP3).

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:

Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
Wearing of closed work clothing is required additionally to the stated personal protection equipment.

## 9. Physical and Chemical Properties

### Information on basic physical and chemical properties

| | |
|---|---|
| Form: | solid |
| Colour: | brown |
| Odour: | odourless |
| pH value: | approx. 5,0<br>(20 °C) |
| Melting temperature: | not applicable |
| boiling temperature: | not applicable |
| Flash point: | not applicable |
| Lower explosion limit: | For solids not relevant for classification and labelling. |
| Upper explosion limit: | For solids not relevant for classification and labelling. |
| Vapour pressure: | (20 °C)<br>negligible |
| Relative vapour density (air): | The product is a non-volatile solid. |
| Solubility in water: | immiscible |
| Self ignition: | not self-igniting |
| Thermal decomposition: | Not a substance liable to self-decomposition according to UN transport regulations, class 4.1. No decomposition if correctly stored and handled. |
| Viscosity, dynamic: | not applicable, the product is a solid |
| Viscosity, kinematic: | not applicable, the product is a solid |
| Explosion hazard: | Based on the chemical structure there is no indication of explosive properties. |
| Fire promoting properties: | not fire-propagating |
| SADT: | Not a substance liable to self-decomposition according to UN transport regulations, class 4.1. |

### Other information

| | |
|---|---|
| Bulk density: | approx. 1.100 kg/m3 |

## 10. Stability and Reactivity

### Reactivity
No hazardous reactions if stored and handled as prescribed/indicated.

Corrosion to metals: No corrosive effect on metal.

### Chemical stability
The product is stable if stored and handled as prescribed/indicated.

### Possibility of hazardous reactions
No hazardous reactions if stored and handled as prescribed/indicated.

### Conditions to avoid
See SDS section 7 - Handling and storage.

### Incompatible materials

Substances to avoid:
None known during use and storage if used according to instructions.

### Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

## 11. Toxicological Information

### Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Of low toxicity after single ingestion. The product has not been tested. The statement has been derived from the properties of the individual components.

The following percentage of the mixture consists of components(s) with unknown hazards regarding the acute toxicity: 17 - 26 %

The following percentage of the mixture consists of components(s) with unknown hazards regarding the acute toxicity: 12 - 19 %

The following percentage of the mixture consists of components(s) with unknown hazards regarding the acute toxicity: 46 - 67 %

Irritation

Assessment of irritating effects:
Eye contact causes irritation. The product has not been tested. The statement has been derived from the properties of the individual components.

Respiratory/Skin sensitization

Assessment of sensitization:
The substance may cause sensitization of the respiratory tract. Sensitization after skin contact possible. The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: Cobalt molybdate*
*Assessment of sensitization:*
*Sensitization after skin contact possible. The substance may cause sensitization of the respiratory tract. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.*
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Germ cell mutagenicity

Assessment of mutagenicity:
Mutagenic properties can not be excluded on the basis of experimental data. The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: Cobalt molybdate*
*Assessment of mutagenicity:*
*The substance was not mutagenic in bacteria. The substance was mutagenic in a mammalian cell culture test system. The substance was mutagenic in studies with mammals. The product has not been fully tested. The statements have been derived in parts from products of a similar structure or composition.*
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Carcinogenicity

Assessment of carcinogenicity:
The substance caused cancer in animal studies. The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: Cobalt molybdate*
*Assessment of carcinogenicity:*
*The substance caused cancer in animal studies. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.*
---

Reproductive toxicity

Assessment of reproduction toxicity:
Causes impairment of fertility in laboratory animals. The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: Cobalt molybdate*
*Assessment of reproduction toxicity:*
*Causes impairment of fertility in laboratory animals. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.*
---

Developmental toxicity

Assessment of teratogenicity:

Indications of possible developmental toxicity/teratogenicity were seen in animal studies. The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: Cobalt molybdate*
*Assessment of teratogenicity:*
*The results of animal studies gave indication of a developmental toxic effects. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.*
---

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: Molybdenum trioxide*
*Assessment of repeated dose toxicity:*
*The substance may cause damage to the kidney after repeated ingestion of high doses, as shown in animal studies.*
---

Aspiration hazard

No aspiration hazard expected.
The product has not been tested. The statement has been derived from the properties of the individual components.

Other relevant toxicity information

Misuse can be harmful to health.

## 12. Ecological Information

### Toxicity

Assessment of aquatic toxicity:
Toxic to aquatic life. Toxic to aquatic life with long lasting effects.
The product has not been tested. The statement has been derived from the properties of the individual components.

### Persistence and degradability

Assessment biodegradation and elimination (H2O):

Not applicable for inorganic substances.

### Bioaccumulative potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: Cobalt molybdate*
*Assessment bioaccumulation potential:*
*Significant accumulation in organisms is not to be expected.*
---

### Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: The product has not been tested. The statement has been derived from the properties of the individual components.

*Information on: Cobalt molybdate*
*Assessment transport between environmental compartments:*
*Volatility: The substance will not evaporate into the atmosphere from the water surface.*
*Adsorption in soil: Adsorption to solid soil phase is possible.The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.*
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### Other adverse effects

The product does not contain substances that are listed in the Montreal Protocol on substances that deplete the ozone layer.

### Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

## 13. Disposal Considerations

### Waste treatment methods

Dispose of in accordance with national, state and local regulations.
Used catalysts may have different hazardous properties than the original products.

Contaminated packaging:
Disposal must be made according to official regulations.

## 14. Transport Information

**Land transport**

ADR

| | |
|---|---|
| UN number or ID number: | UN3077 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (COBALT MOLYBDATE) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

RID

| | |
|---|---|
| UN number or ID number: | UN3077 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (COBALT MOLYBDATE) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

**Inland waterway transport**

ADN

| | |
|---|---|
| UN number or ID number: | UN3077 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (COBALT MOLYBDATE) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

Transport in inland waterway vessel
Not evaluated

**Sea transport**

IMDG

| | |
|---|---|
| UN number or ID number: | UN 3077 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (COBALT MOLYBDATE) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes<br>Marine pollutant: YES |

| | |
|---|---|
| Special precautions for user: | EmS: F-A; S-F |

**Air transport**

IATA/ICAO

| | |
|---|---|
| UN number or ID number: | UN 3077 |
| UN proper shipping name: | ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S. (COBALT MOLYBDATE) |
| Transport hazard class(es): | 9, EHSM |
| Packing group: | III |
| Environmental hazards: | yes |
| Special precautions for user: | None known |

### Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

## 15. Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

## 16. Other Information

This product is of industrial quality and unless otherwise specified or agreed intended exclusively for industrial use.

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

| | |
|---|---|
| Acute Tox. | Acute toxicity |
| Eye Dam./Irrit. | Serious eye damage/eye irritation |
| Resp. Sens. | Respiratory sensitization |
| Skin Sens. | Skin sensitization |
| Muta. | Germ cell mutagenicity |
| Carc. | Carcinogenicity |
| Repr. | Reproductive toxicity |
| Aquatic Acute | Hazardous to the aquatic environment - acute |
| Aquatic Chronic | Hazardous to the aquatic environment - chronic |
| STOT SE | Specific target organ toxicity — single exposure |
| Skin Corr./Irrit. | Skin corrosion/irritation |
| H303 | May be harmful if swallowed. |
| H334 | May cause allergy or asthma symptoms or breathing difficulties if inhaled. |
| H317 | May cause an allergic skin reaction. |
| H341 | Suspected of causing genetic defects. |
| H350 | May cause cancer by inhalation. |

| | |
|---|---|
| H360 | May damage fertility. Suspected of damaging the unborn child. |
| H401 | Toxic to aquatic life. |
| H411 | Toxic to aquatic life with long lasting effects. |
| H319 | Causes serious eye irritation. |
| H335 | May cause respiratory irritation. |
| H351 | Suspected of causing cancer. |
| H412 | Harmful to aquatic life with long lasting effects. |
| H314 | Causes severe skin burns and eye damage. |

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.